

Homework

Basics in linear algebra and numerical methods, 2025/26. semester I.

General instructions: Each week, you have to submit the solution of three arbitrary exercises. This should be done as a Python code. Comments should be written in English.

Scores: 3×3 per week; altogether: 54 in the whole semester.

Week 3

- (a) For the matrix $A = \begin{pmatrix} 1 & 0 & 4 \\ 5 & 2 & -2 \\ 2 & 9 & 3 \end{pmatrix}$ the conditions for our iterative algorithm for solving $A\mathbf{x} = \mathbf{b}$

are not satisfied. But taking $PA\mathbf{x} = P\mathbf{b}$ can give such a matrix $\hat{A} = PA$ on the left hand side that does satisfy our conditions. Give such a matrix P . (This is exceptional : not a Python exercise.)

- (b) Use vectorized operations to compute the vector

$$\left[A[1, 1] - \sum_{k \neq 1} |A[1, k]|, A[2, 2] - \sum_{k \neq 2} |A[2, k]|, \dots, A[N, N] - \sum_{k \neq N} |A[N, k]| \right]$$

for a given square matrix $A \in \mathbb{R}^{N \times N}$ (it is not necessary to write a function just rather test it on a matrix).

- (c) Take a random matrix $A \in \mathbb{R}^{11 \times 10}$ and a random vector $\mathbf{b} \in \mathbb{R}^{11}$. „Solve" the overdetermined system $A\mathbf{x} = \mathbf{b}$ to have $\mathbf{x}_h$ and compute the error $\|A\mathbf{x}_h - \mathbf{b}\|$ of the approximation. You may use the subroutine `scipy.linalg.lstsq`.